

Section-A

$$\textcircled{1} \div 2 \Rightarrow \frac{14}{2} = \frac{a + 2b}{2}$$

$$\Rightarrow \frac{36}{22} = \frac{30a + 2b}{22}$$

Al.

(i) d) 20

(ii) d) 55

(iii) Soln

eqn: $y = ax + b$

The normal equations are:

$\sum y = a \sum x + nb$

$\sum xy = a \sum x^2 + b \sum x$

x	y	xy	x ²
-4	4	-16	16
1	6	6	1
2	8	16	4
3	10	30	9
2	28	36	30

$\sum x = 2 \quad \sum xy = 36$

$\sum y = 28 \quad \sum x^2 = 30$

$28 = 2a + 4b \quad \text{--- (1)}$

$36 = 30a + 2b \quad \text{--- (2)}$

$a = 0.758$

$b = 6.620$

$\therefore y = 0.758x + 6.620$

(iv) Soln

eqn $\Rightarrow y = ax + b$

The normal equation are

$\sum y = a \sum x + nb$

$\sum xy = a \sum x^2 + b \sum x$

x	y	xy	x ²
0	1	0	0
1	1.8	1.8	1
2	3.3	6.6	4
3	4.5	13.5	9
4	6.3	25.2	16
10	16.9	47.1	30

$\sum x = 10 \quad \sum xy = 47.1$

$\sum y = 16.9 \quad \sum x^2 = 30$

$$16.9 = 10a + 5b \quad \text{--- (1)}$$

$$47.1 = 30a + 10b \quad \text{--- (2)}$$

On solving (1) & (2).

$$a = 1.33$$

$$b = 0.72$$

$$y = 1.33x + 0.72.$$

Q2.

(i) a) 6

(ii) c) 38

(iii) gn

$$T_1 = [4, 1] A$$

$$T_2 = [-2, 2] A.$$

$$I = a_1 T_1 + b T_2.$$

$$I = [2, 8] A.$$

Soln

$$[2, 8] = a[4, 1] + b[-2, 2]$$

$$[2, 8] = [4a, a] + [-2b, 2b]$$

$$[2, 8] = [4a - 2b, a + 2b]$$

$$4a - 2b = 2 \quad \text{--- (1)}$$

$$a + 2b = 8 \quad \text{--- (2)}$$

On solving (1) & (2)

$$a = 2$$

$$b = 3.$$

$$\therefore I = 2T_1 + 3T_2$$

(iv)

$$A = \begin{bmatrix} 0 & -2 & -2 \\ -1 & 1 & 2 \\ -1 & -1 & 2 \end{bmatrix}$$

The characteristic equation is
 $\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0.$

$$S_1 = 0 + 1 + 2$$

$$\boxed{S_1 = 3}$$

$$S_2 = \begin{vmatrix} 1 & 2 \\ -1 & 2 \end{vmatrix} + \begin{vmatrix} 0 & -2 \\ -1 & 2 \end{vmatrix} + \begin{vmatrix} 0 & -2 \\ -1 & 1 \end{vmatrix}$$

$$= 4 + 0 + (-2) + -2$$

$$\boxed{S_2 = 0.}$$

$$S_3 = |A| = -4$$

$$\lambda^3 - 3\lambda^2 + 0\lambda + 4 = 0.$$

$$\lambda_1 = -1$$

$$\lambda_2 = 2$$

$$\lambda_3 = 2.$$

$$(A - \lambda I)X = 0$$

$$\lambda = -1$$

$$\begin{bmatrix} 0 & -2 & -2 \\ -1 & 1 & 2 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & -2 & -2 \\ -1 & 2 & 2 \\ -1 & -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\begin{bmatrix} -1 & 2 & 2 & -1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -1 & 3 & -1 & -1 \end{bmatrix}$$

$$\frac{x}{8} = \frac{y}{1} = \frac{z}{3}$$

$$\begin{vmatrix} 2 & 2 \\ -1 & 3 \end{vmatrix} \begin{vmatrix} 2 & -1 \\ 3 & -1 \end{vmatrix} \begin{vmatrix} -1 & 2 \\ -1 & -1 \end{vmatrix}$$

$$\frac{x}{8} = \frac{y}{1} = \frac{z}{3}$$

$$X_1 = \begin{bmatrix} 8 \\ 1 \\ 3 \end{bmatrix}$$

$$\lambda = 2$$

$$\begin{bmatrix} -2 & -2 & -2 \\ -1 & -1 & 2 \\ -1 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\begin{bmatrix} -2 & -2 & -2 & -2 & -2 \\ -1 & -1 & 2 & -1 & -1 \end{bmatrix}$$

$$\frac{x}{-6} = \frac{y}{6} = \frac{z}{0}$$

$$\frac{x}{-6} = \frac{y}{6} = \frac{z}{0}$$

$$X_2 = \begin{bmatrix} -6 \\ 6 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Since this is a symmetric matrix, if the eigen values are repeated then, $X_2 = X_3$

$$X_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

The largest eigen value is 2

$$X_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Here 1 is the highest value in eigen vector and the corresponding city is city B. $\therefore$ Transportation demand flow is highest among the most significant Eigen vector. Here.

City B followed by City C followed by City A.

B1.

(i) a) 1955

(ii) a) Upward.

(iii) Given

$$\sum x = 0 \quad \sum y = 6$$

$$\sum x^2 = 28 \quad \sum xy = 40$$

$$\sum x^3 = 0 \quad \sum x^2y = 6$$

$$\sum x^4 = 196$$

The normal equations are

$$6 = 28a + 0b + 7c$$

$$40 = 0a + 28b + 0c$$

$$6 = 196a + 0b + 28c$$

$$a = -0.219$$

$$b = 1.428$$

$$c = 1.714$$

$$y = -0.219x^2 + 1.428x + 1.714$$

(iv) given

$$p(x) = -2x^2 + 80x - 600$$

(a) x-intercept (y=0)

$$0 = -2x^2 + 80x - 600$$

$$\boxed{x_1 = 10} \quad , \quad \boxed{x_2 = 30}$$

y-intercept (x=0)

$$y = -2(0) + 80(0) - 600$$

$$\boxed{y = -600}$$

(b) Vertex (h, k)

$$h = \frac{-b}{2a} = \frac{+80}{2(-2)} = -20$$

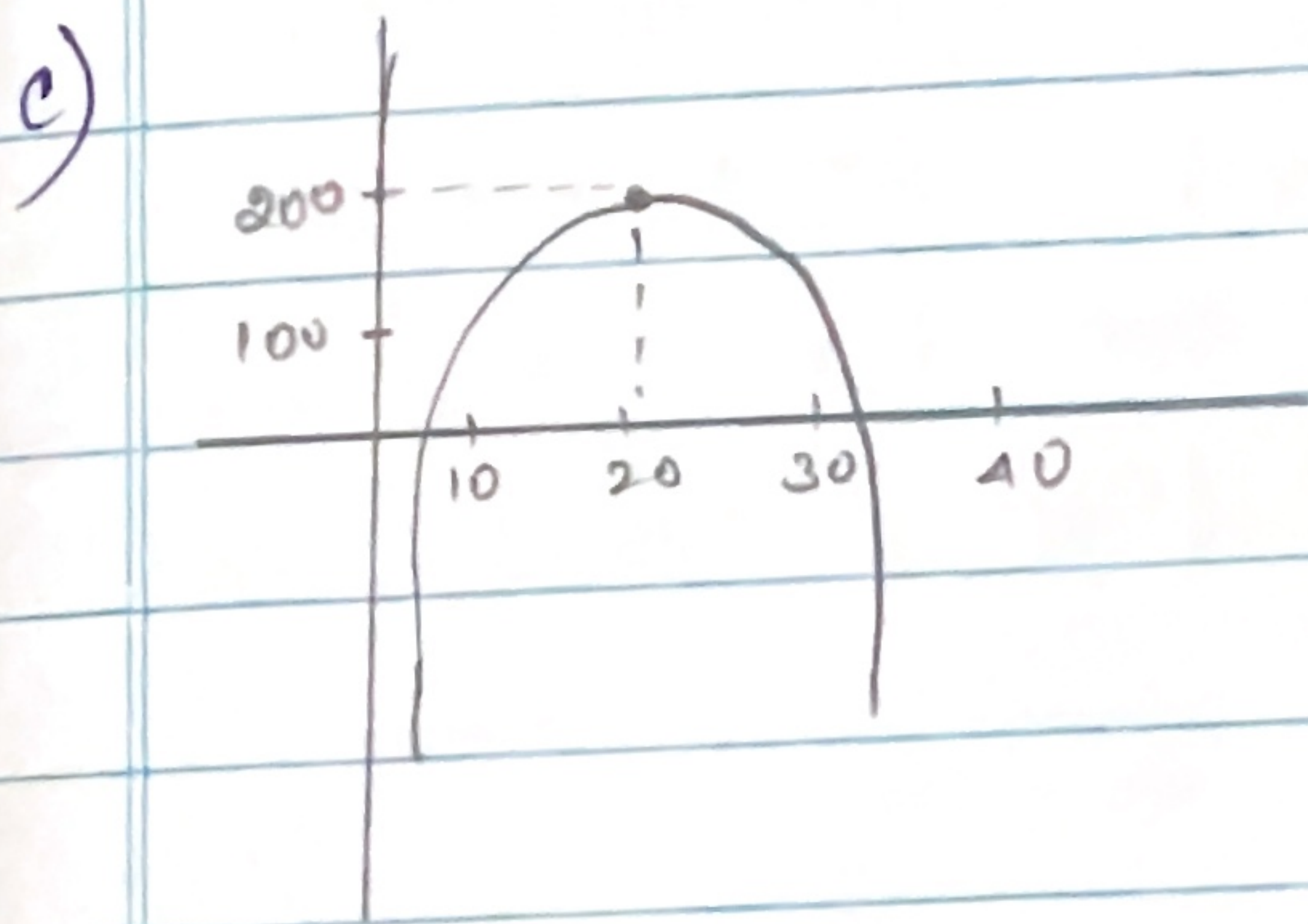
$$k = f(00)$$

$$\begin{aligned}
 f(20) &= -2(20)^2 + 80(20) - 600 \\
 &= -800 + 1600 - 600 \\
 &= 200.
 \end{aligned}$$

vertex $(20, 200)$.

From this it means.

20 is the maximum no. of CD's produced and 200 is the maximum ^{profit} ~~cost~~ yield / earned.



a) $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

b) The characteristic eqn is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$

$$S_1 \Rightarrow 12$$

$$S_2 \Rightarrow \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} + \begin{vmatrix} 6 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 6 & -2 \\ -2 & 3 \end{vmatrix}$$

$$\Rightarrow 8 + 14 + 14$$

$$S_2 \Rightarrow 36$$

$$S_3 \Rightarrow 32 \Rightarrow |A|.$$

$$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0.$$

$$\lambda_1 = 8, \lambda_2 = 2, \lambda_3 = 2$$

All three eigen value are positive
then Nature of the motion is
stable motion

Nature $\Rightarrow$ positive definite.

(B3)

(i) d) Partial definite

(ii) a) A^T

(iii) Given

$$6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$$

(iv)

a) $\lambda = 2$

$$(A - \lambda I) X = 0$$

$$\begin{bmatrix} -2 & 2 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$-4x + 2y = 0$$

$$2x - y = 0 \quad \text{or } 2x - y = 0$$

$$y = 0$$

$$-y = 0$$

$$y = 0$$

$$2x - y = 0$$

$$x = 0$$

$$2x = 0$$

$$x = 0$$

It is symmetric, by the property of orthogonal.

$$X_3 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$X_2 = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$X_3^T \cdot X_2 = 0$$

$$\begin{bmatrix} -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

C2

(i) b) 3

(ii)	t	H	log H	log t (x)	xy	x ²
	2	5	0.6989	0.3010	0.2103	0.0906
	4	9	0.9542	0.6020	0.5744	0.3624
	6	15	1.1760	0.7781	0.9150	0.6054
	8	24	1.3802	0.9030	1.2463	0.8154
	10	35	1.5440	1	1.5440	1
	80	88	5.7533	3.5841	4.4900	2.8738

$$5.7533 = 5A + 3.5841b$$

$$4.4900 = 3.5841A + 2.8738b$$

$$A = 0.2896 \quad b = 1.2011$$

$$\log a = 0.2896 \Rightarrow a = 10^{0.2896} \Rightarrow 1.9480$$

$$a = 1.9480 \quad b = 1.2011$$

C3

~~Y increases linearly~~

(i) d) y is directly proportional to x.

(ii) given.

$$5A + 2.079b = 12.563$$

$$2.079A + 1.169b = 5.623$$

$$A \Rightarrow 1.967 \quad b \Rightarrow 1.311$$

$$\log a = 1.967 \quad a = 10^{1.967}$$

$$a = 92.682$$

$$b = 1.311$$

(B2)

(i) b)
$$\begin{bmatrix} 2 & -3 \\ -3 & 3 \end{bmatrix}$$

(ii) c) 2

(iii)
$$A = \begin{bmatrix} 3 & 2 & 1 \\ 0 & -1 & 4 \\ 0 & 0 & 2 \end{bmatrix}$$

a) $\lambda_1 = 3, \lambda_2 = 2, \lambda_3 = -1$

b) $3x^2 - y^2 + 2z^2$

c) rank $\Rightarrow 3$

nature $\Rightarrow$ indefinitestability $\Rightarrow$ unstable.

(iv)

a)
$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$s_1 \Rightarrow 0$$

$$s_2 = -1 + 1 + 1$$

$$s_2 \Rightarrow 1$$

$$s_3 \Rightarrow 2$$

$$\lambda^3 - 0\lambda^2 + 1\lambda - 2 = 0$$

$$\lambda_1 = 1, \text{ ~~1, 1~~$$

$$\lambda_2 = -0.5, \lambda_3 = -0.5$$

Nature of the motion

 $\Rightarrow$ indefinite

Complex situation with trade offs and the potential for saddle points, where the must carefully balance inputs